AI Presentation 2026

Introduction and Use Examples

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2. Goals for this Presentation

- Discuss language-based AI models
- Explore how you might use them
- Layout benefits and liabilities, especially Data Centers

3. Why Should You Care About AI?

- You've heard a lot about Artificial Intelligence (AI) in the news and your email newsletters.
- AI is powerful and scary. It can be the agent for much good and much bad.
- AI can also take a lot of the friction out of maintenance tasks for you and your surroundings. It can explain technology, help you with settings, review your medications ...
- It can also be fun to use AI to dig up facts about the movies over dinner.
- You want to know enough to make good voting choices, and good conversation.

5. AI Covers More Than Chatting

Some AI models play chess and go at grand master level.

Others fold proteins and predict weather as well as, or better than, existing methods.

This presentation covers one subset: Large Language Models, or LLMs.

- A Large Language Model is a program trained on text that answers in ordinary language.
- A Chat Bot is a web page or mobile application that lets you ask an LLM a series of questions, with answers for each.

Chess engines, protein folders, and LLM chat bots share AI name and little else.

6. What a Large Language Model (LLM) Does

Answers questions drawn from a large body of text (Chat Bot)

- Much of the web, scientific papers, code, forum sites, and Wikipedia

Helps build documents (CLI)

- Slide decks, PDF documents, and web pages

Carries out tasks (AI Agent)

- Testing code, sorting email, and similar routine work

7. Reference Articles

Benefits of LLM AIs at the global and societal level.docx

Global_Liabilities_of_LLM_AIs.docx

What Are Data Centers Doing? A Mind-Boggling Amount of Math. - The New York Times

6 Takeaways From Ezra Klein's Interview With Bill Gates - The New York Times

Six Charts That Show Just How Much We Need A.I. - The New York Times

8. Beginning Use Examples

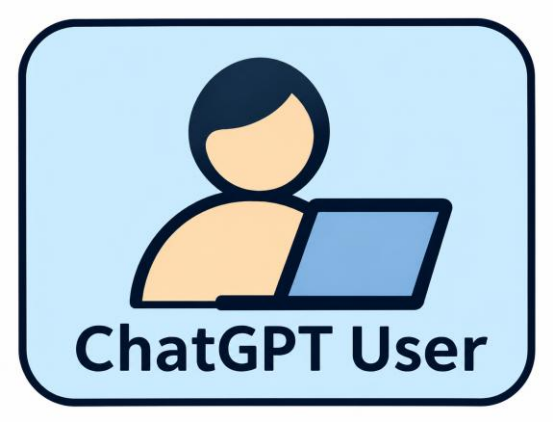
I will open a few of these examples and demonstrate them in a chat bot panel.

Follow along at either site, both free to start:

- ChatGPT.com
- Claude.ai

Four kinds of everyday use:

- a. Everyday practical help
- b. Tech help
- c. Personal connections
- d. Reading recommendations



9. Everyday Practical Help and Tech Help

Everyday practical help

What are the midterm election results for Nov 3, 2026 for a voter in Urbana IL 61802?

Create a mediterranean dinner menu with recipes and ingredients. Favor short ingredient lists.

Explain these medical test results [paste results text here]

Tech help

What is a podcast?

How to set up a video call?

Is this a scam [paste potential scam text here]?

Where in Champaign/Urbana can I ship out used toner cartridge for recycling?

Try these at: ChatGPT.com Claude.ai

10. Reading and Personal Connections

Reading recommendations

- Sci-Fi titles of books similar to those of Iain M. Banks
- New York Times and Atlantic articles on AI
- Create list of biographies of famous people

Personal connections

- What are good activities for visiting friends with children in Champaign-Urbana area?
- How to set up recurring Zoom meeting with friends?
- Draft wedding toast [add details]
- Draft letter to reconnect with old friend [add details]

A draft is a starting point. The sender still writes the letter.

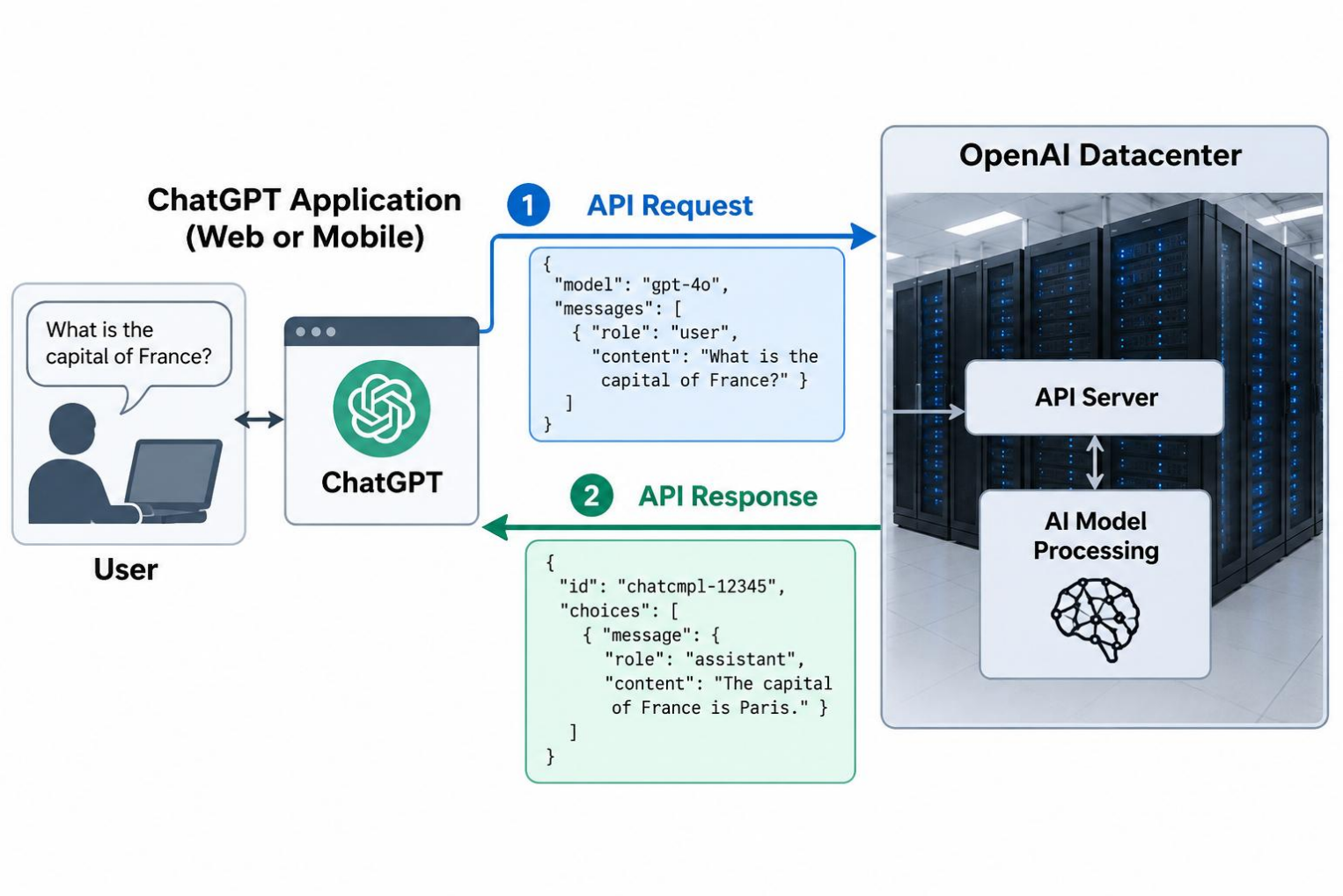
Try these at: ChatGPT.com Claude.ai

11. ClarkLindsey Use of LLM AIs

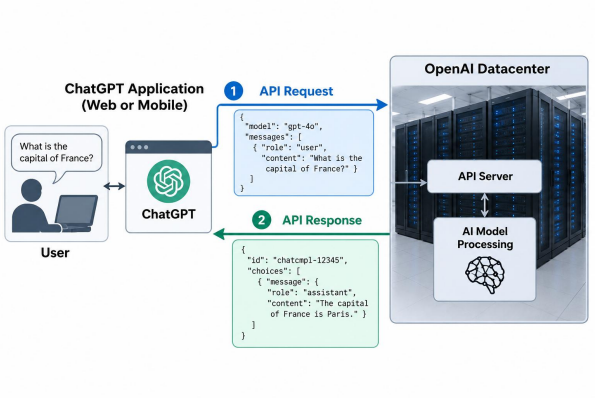
ClarkLindsey Administration Uses Large Language AI

12. How Large Language Model AI Chat Bots Work

The question leaves your device. The answer comes back from a building you will never see.



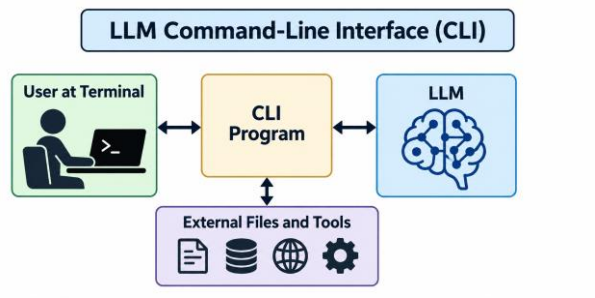
14. Compare Chat Bot, CLI, and Agent



Chat Bot

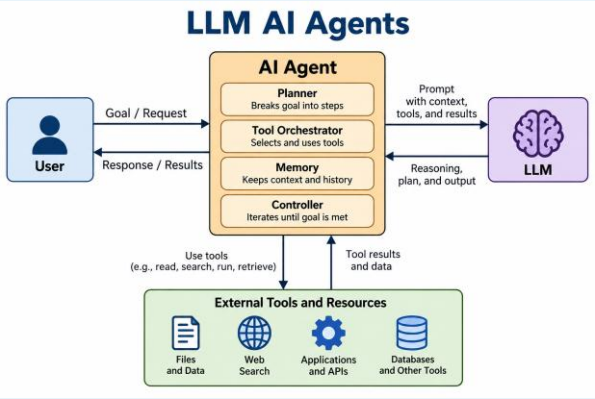
Works well for question-answer conversations.

Awkward for handling multiple files.



Command Line Interface

Content providers use CLI for managing code, documentation, and presentation files.

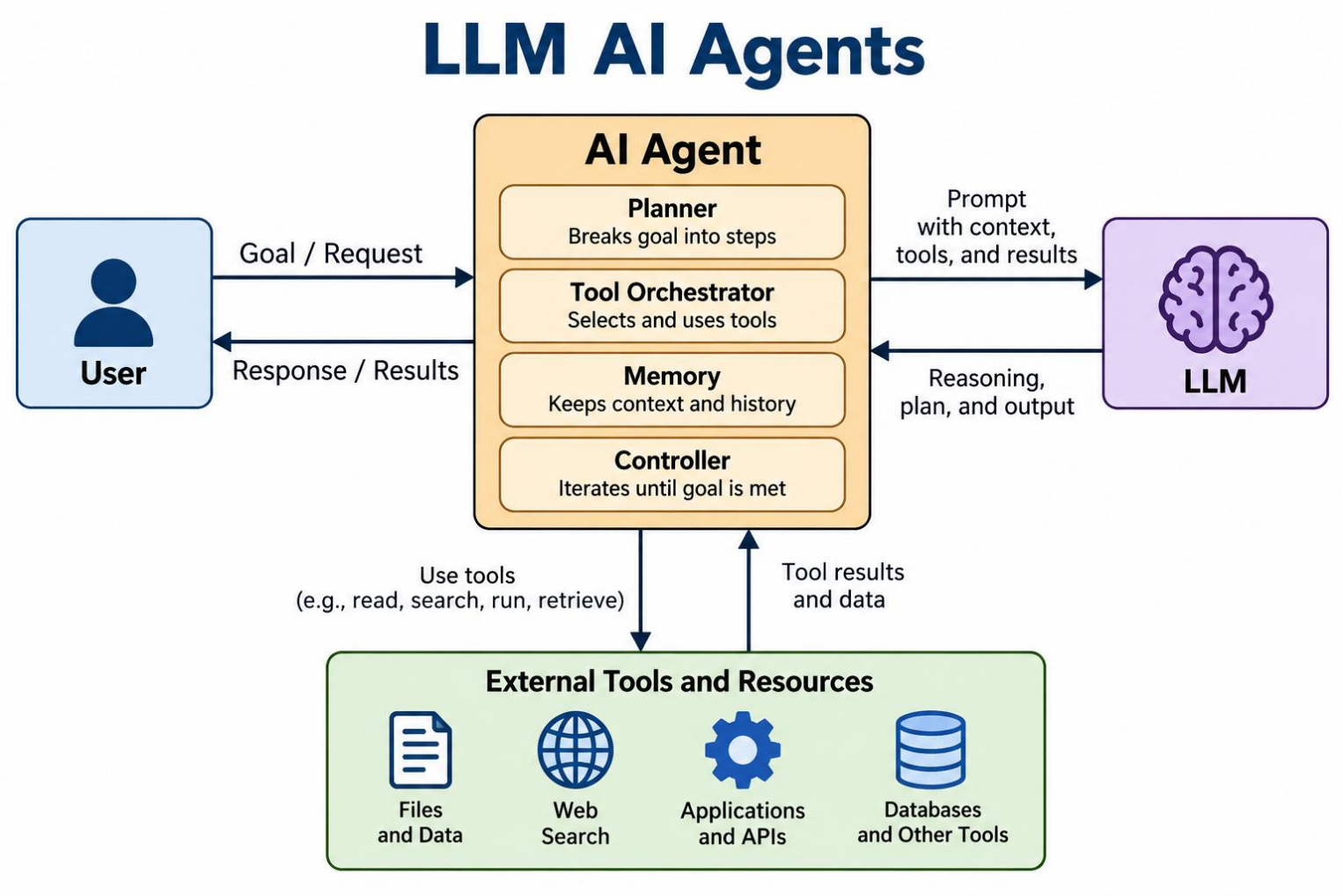


AI Agent

Runs autonomously on complex tasks.

17. Autonomous AI Agents

An agent runs on its own from a starting prompt until it reaches a goal you set.



18. Agents: The Good, the Bad, and the Dangerous

Good news

- When an agent runs well, a lot gets done without your attention.

Bad news

- A misdirected agent produces junk and deletes good content.
- Much of what it does rests on meaning it inferred from your request, so the error is hard to prevent.
- Developers test in a sandbox, meaning an isolated copy, and deploy only once it works.

Really bad news

- Bad actors run swarms of agents against banking security and infrastructure controls, and post misinformation at scale.

20. Seven Areas of Benefit (1 of 2)

Education and knowledge access

- Tutoring at any hour reaches students where teachers are scarce, in hundreds of languages, at each learner's pace.

Healthcare

- Drafting and summarizing return clinician time to patients, and restate instructions in plain language.

Economic opportunity

- A solo operator gets marketing copy, bookkeeping, and customer answers that once required a hire.

Science and research

- Literature review and routine code move faster, and the model translates the jargon between fields.

21. Seven Areas of Benefit (2 of 2)

Inclusion and accessibility

- People with disabilities read, write, and navigate with less friction. Legal basics, tax forms, and government paperwork reach people who cannot pay professional fees.

Public services and crisis response

- Agencies simplify public information, and responders summarize reports across languages while the situation moves.

Language and culture

- Communities document endangered languages, and creative work reaches readers in languages it was never published in.

What the gains depend on: access, accuracy safeguards, and deployment choices.

23. Eight Areas of Liability (1 of 2)

Security and misuse

- Personalized phishing and fraud scale up, and mounting an attack demands less skill than it did.

Environmental and resource costs

- Training and answering both draw large amounts of electricity. Cooling consumes water, and the load strains local grids.

Misinformation and trust

- Fabricated statements arrive in confident wording. The industry calls them hallucinations. Trust in genuine material erodes, since anyone can now dismiss a real document as AI-generated.

Economic disruption

- Writing, support, translation, programming, and administration face job losses or wage pressure, unevenly distributed.

24. Eight Areas of Liability (2 of 2)

Inequality and access

- The models serve the connected first and perform worse in lower-resource languages.

Bias and fairness

- Models restate the stereotypes in their training text, which matters when institutions use them for hiring, lending, or benefits.

Privacy and data rights

- Text typed into a prompt gets stored. Training used copyrighted and personal material without clear consent, and the courts have not settled ownership.

Human and social effects, governance and accountability

- Skills people stop exercising weaken. When a model causes harm, no one has settled who answers for it.

Safeguards, regulation, transparency, and oversight each cut these risks. How much depends on deployment.

25. Data Centers: Power Is the Unit

Every answer a chatbot gives arrives from a building that holds racks of computers, the power to run them, and the cooling to carry away their heat.

Capacity gets measured in power, not in floor space, because power binds:

- A megawatt (MW) is one million watts.
- A 100 MW campus draws about what 80,000 American homes draw on average.
- A gigawatt (GW) is one thousand megawatts, roughly one large nuclear reactor.

Treat every percentage that follows as an estimate. Sources disagree, and construction outruns the tables.

26. Sixty Years, Ten Thousand Times the Power

Operators site these buildings for power, fibre, workforce, tax terms, and a cool climate.

Era	Dominant driver	Scale
1960s-1980s	Mainframe computing	10-100 kW
1990s	Client/server and early web hosting	100 kW to 1 MW
2000s	Dot-com boom, search, early SaaS	1-10 MW per facility
2010s	Cloud, mobile, video streaming	10-200 MW per campus
2020-2024	Pandemic cloud migration, early AI	100-500 MW per campus
2025+	Large-scale AI training and inference	500 MW to 1+ GW per campus

27. Where the Buildings Stand

Operators follow power, fibre, workforce, political stability, tax terms, and cool air.

Region	Share	Key concentrations
United States	~35%	Northern Virginia (Ashburn) is the largest market on earth; also Dallas, Silicon Valley, Chicago, Phoenix, Atlanta
European Union and UK	~22%	Amsterdam, Frankfurt, Dublin, London, Paris; the Nordics add cold climate and renewable power
China	~15%	Beijing, Shanghai, Shenzhen, Guangzhou; new capacity moves west to Guizhou and Inner Mongolia
India and Australia	~7%	Mumbai, Chennai, Hyderabad, Pune grow fastest; Sydney and Melbourne serve the Pacific
Southeast Asia	~6%	Singapore is the hub; land and power limits push growth to Johor Bahru, Jakarta, Bangkok
Rest of world	~6%	UAE and Saudi Arabia invest in AI-scale campuses; also South Africa, Japan, South Korea
Canada and Mexico	~4%	Toronto, Montreal (cheap hydro), Vancouver; Queretaro, Mexico City, Monterrey
Central and South America	~3%	Brazil (Sao Paulo) dominates, with growth in Chile, Colombia, and Argentina
Russia	~2%	Moscow dominates; law requires Russian user data to stay on Russian soil

28. What the Machines Actually Do

Approximate shares of global data center computing, 2024 and 2025.

Category	Share	What runs there
Enterprise and business IT	~25%	Accounting, sales, email, internal applications
AI and machine learning	~18%	Training models and answering queries; up from 10% in 2022
Video streaming	~15%	Netflix, YouTube, TikTok: storage and format conversion
Social media	~8%	Feed ranking, content storage, ad targeting
E-commerce and search	~11%	Transactions, logistics, crawling, indexing
Financial services	~5%	Trading, fraud detection, payments, reporting
Everything else	~18%	Communications, gaming, science, government, health, security

30. Financials: OpenAI, 2022 to 2025

A nonprofit with a billion dollars in pledges reached 13 billion in revenue ten years later.

Year	Revenue	Compute	Operating loss	Net loss
2022	~$28M	n/d	n/d	~$0.54B
2023	~$1B est.	n/d	n/d	n/d
2024	$3.7B	~$7B est.	$8.78B	~$5B
2025	$13.07B	~$17.2B	**$20.92B**	$38.53B

31. Financials: Anthropic, 2022 to 2025

Started five years later, at a smaller scale, and shows the same pattern.

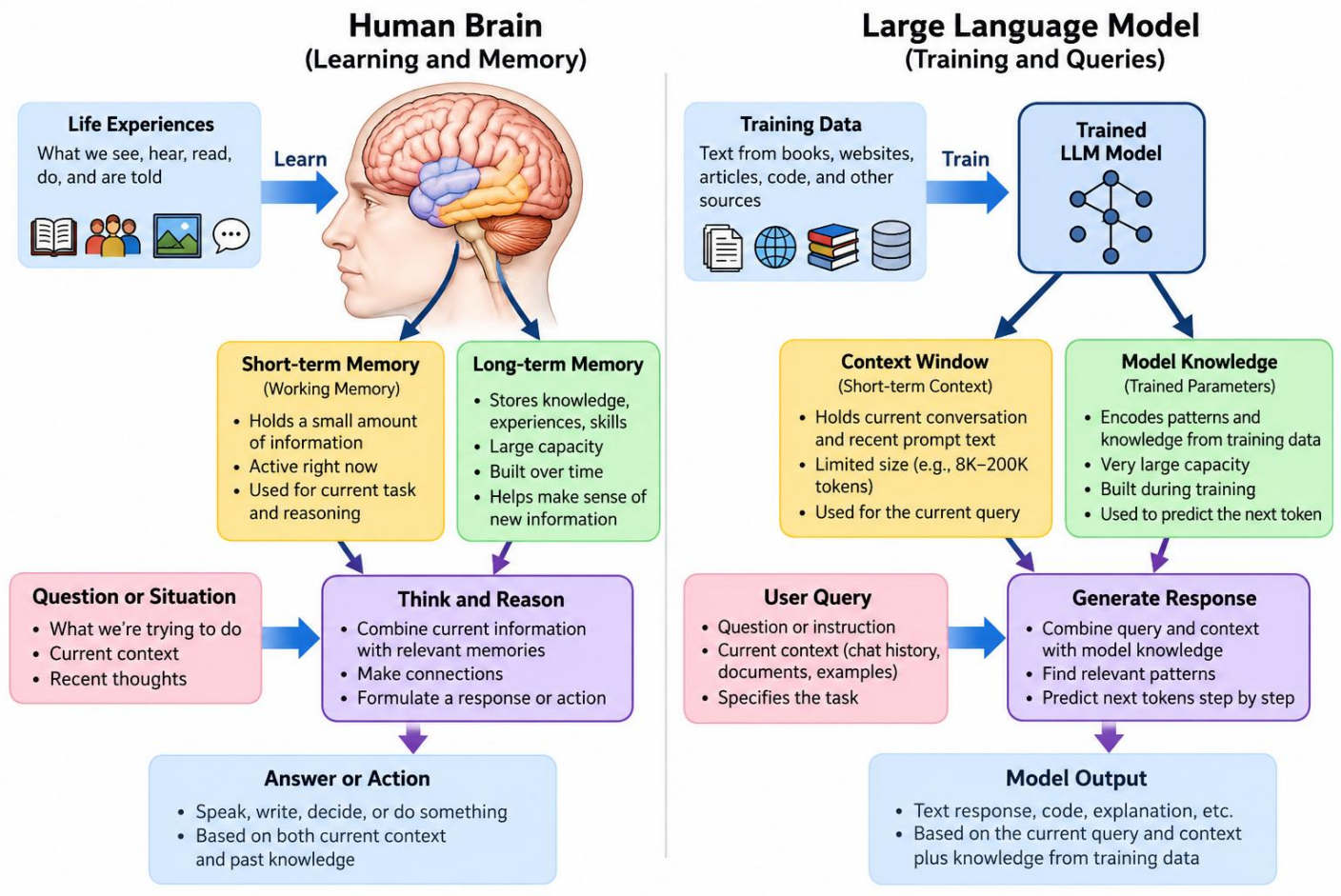
Year	Revenue	Compute	Operating loss	Net loss
2022	~$10M est.	n/d	n/d	n/d
2023	~$100M	n/d	n/d	n/d
2024	$386M	~$2.5B	$2.98B	$8.31B
2025	$4.59B	$7.33B	**$8.06B**	$41.97B

32. A Thought Experiment

- On the next page find an analogy between human thought processing and LLM processing
  - Functions are similar
  - Work flow is very different

33. Human Brain and LLM Differences, Side by Side Analogy

The same path runs down both columns: learn, store, recall, answer.



34. Significant Differences

The human brain writes parts of its short term memory into long term memory.

LLM models do not do that.

> Letting query context modify trained data would open the door to corrupting the training data with bad information, by accident or on purpose.

That corruption may happen with humans too. Perhaps that's why we have cults and misinformation.

In an LLM Chat Bot conversation, all of the context travels with every request. The remote model stores nothing between requests.

35. Large Language Model AI - Take Aways

Chat Bots provide useful answers - just ask the questions

- Technology, health care, things to do

LLM AI Technologies have big **benefits**

- Help students, small businesses, people with disabilities; free time from routine

They also have big **liabilities**

- Big environmental impact: power, water, e-waste
- More effective scams and infrastructure attacks

What you can do

- Read AI articles and share them
- Talk about AI over dinner
- Vote with purpose

That’s All Folks!

Thanks for Coming